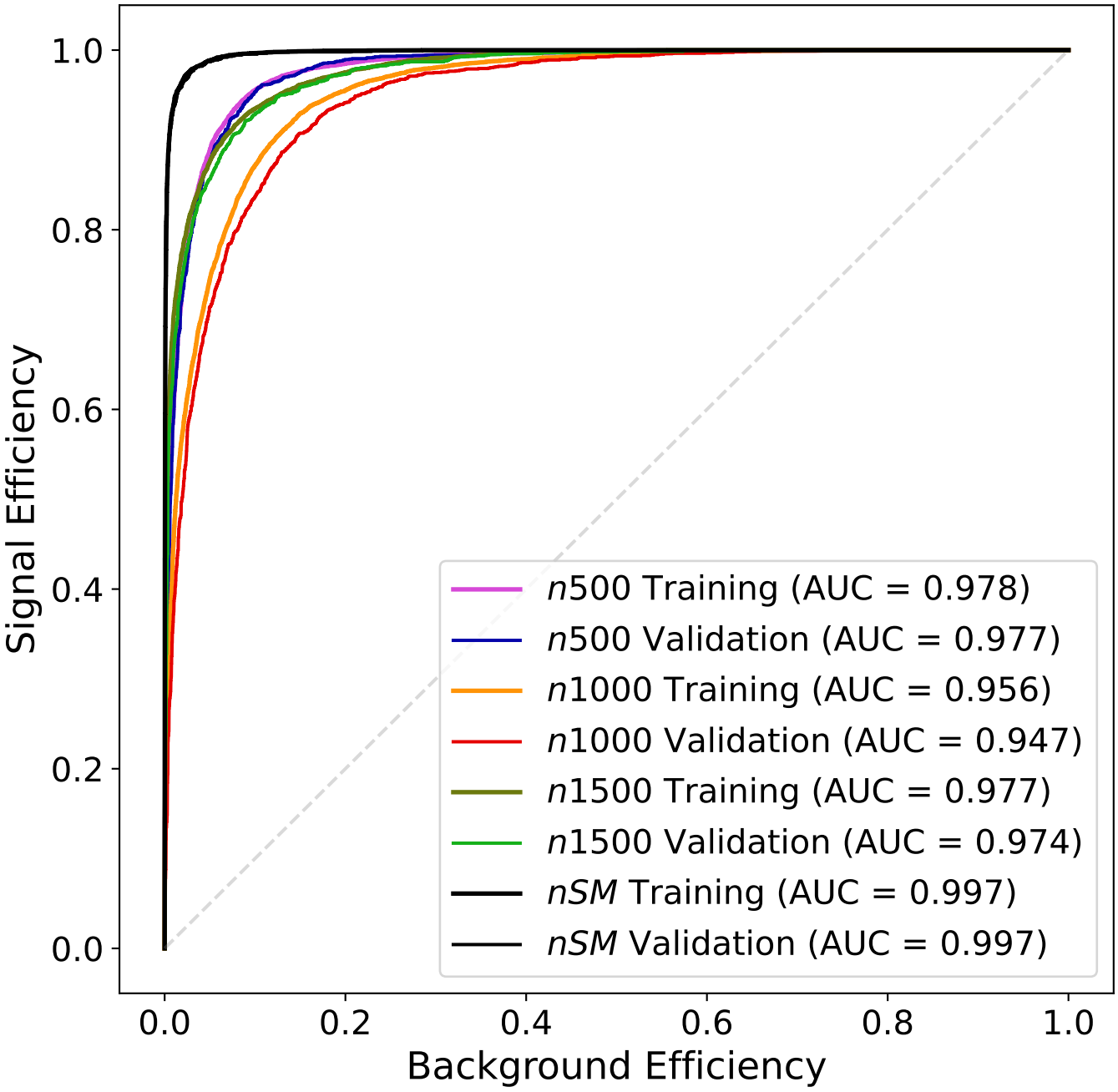
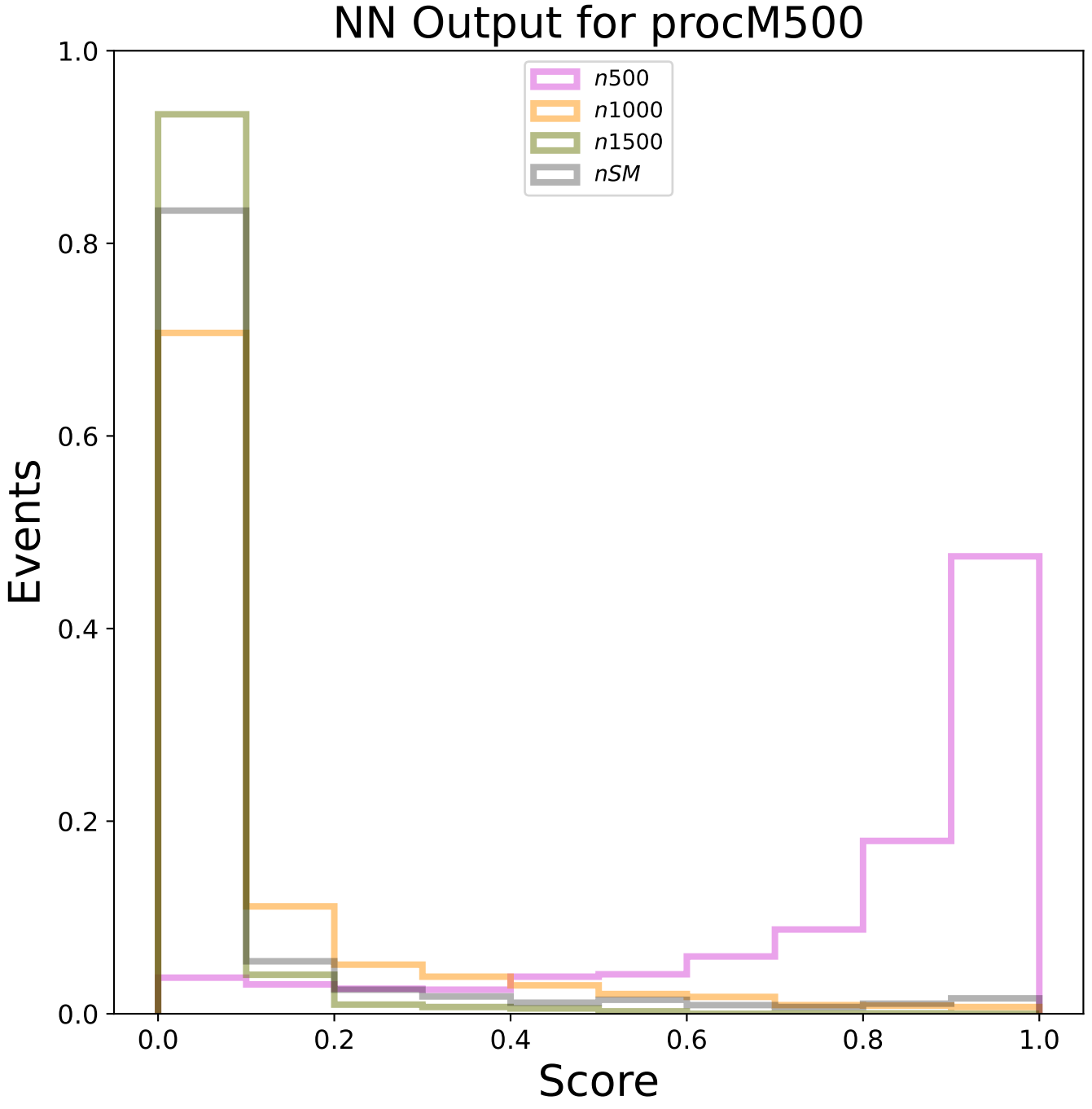
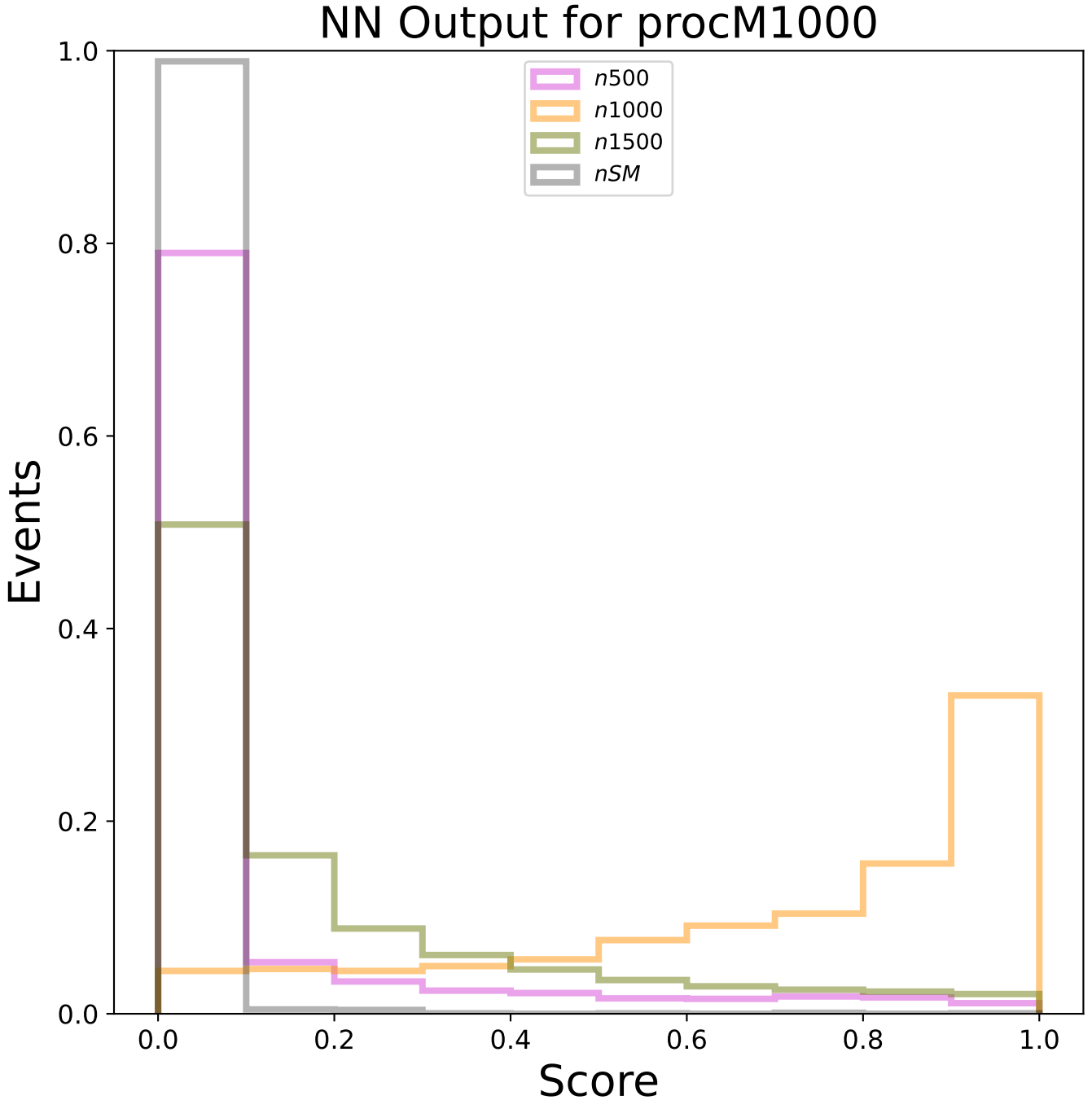
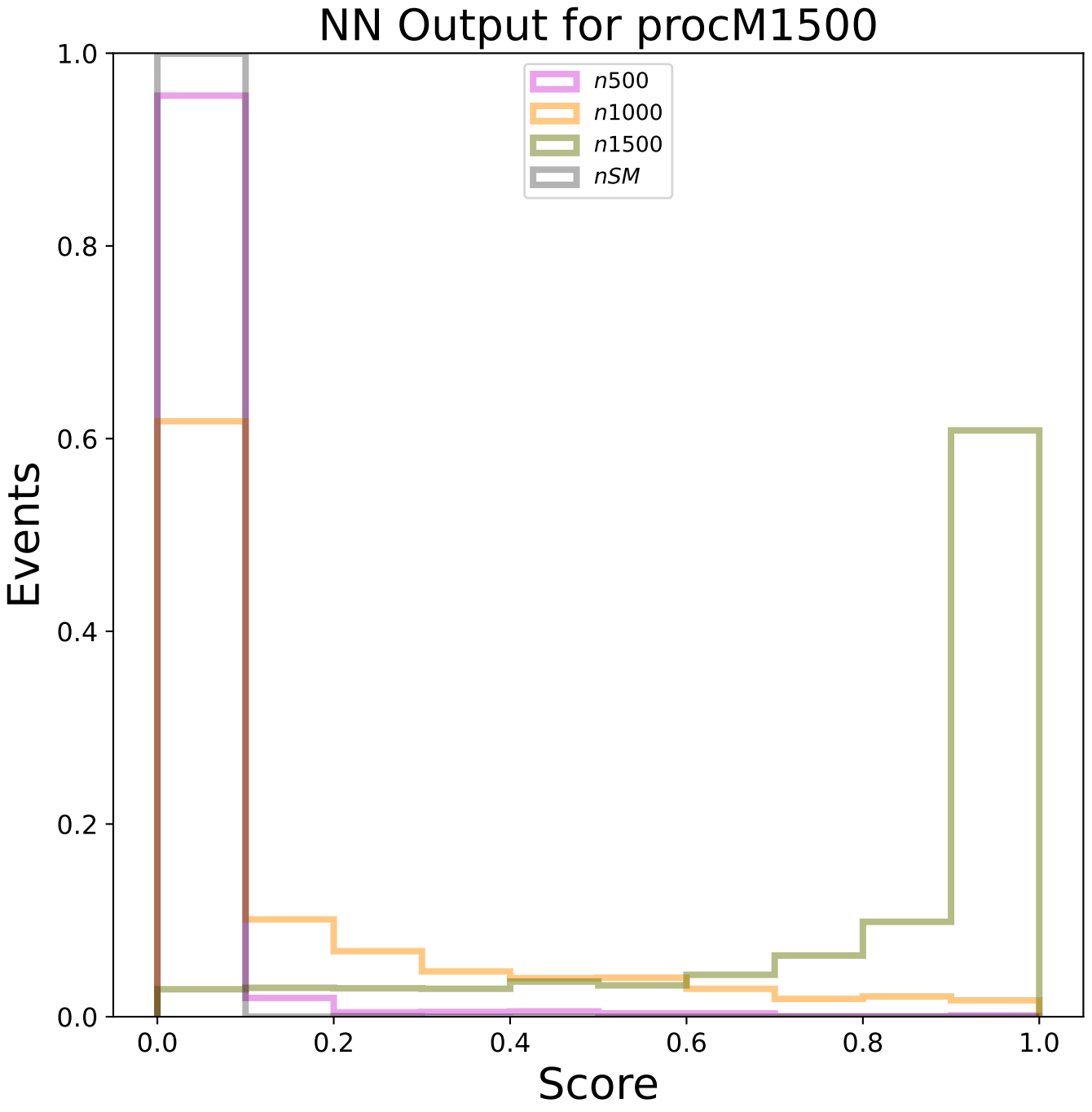
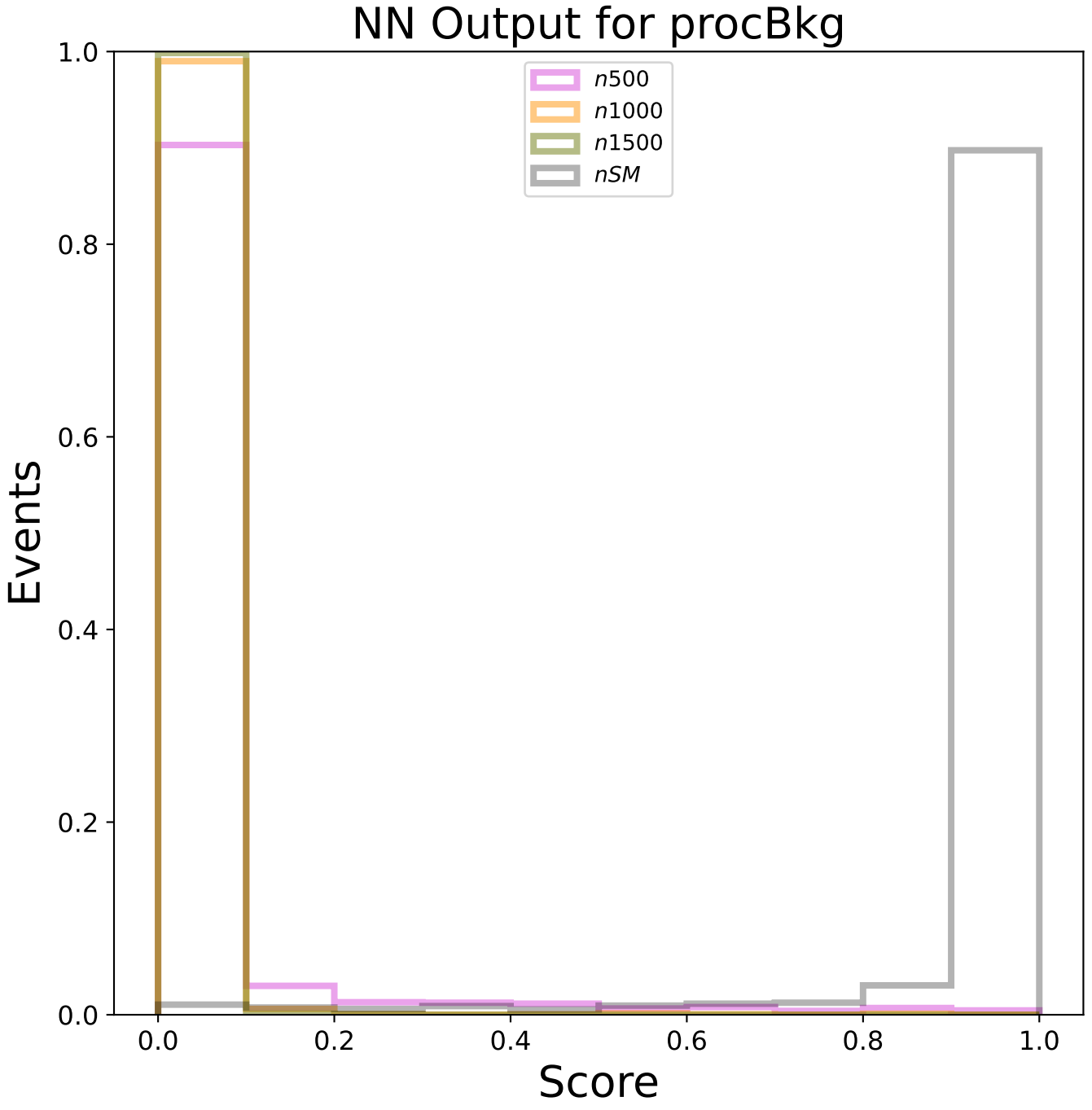
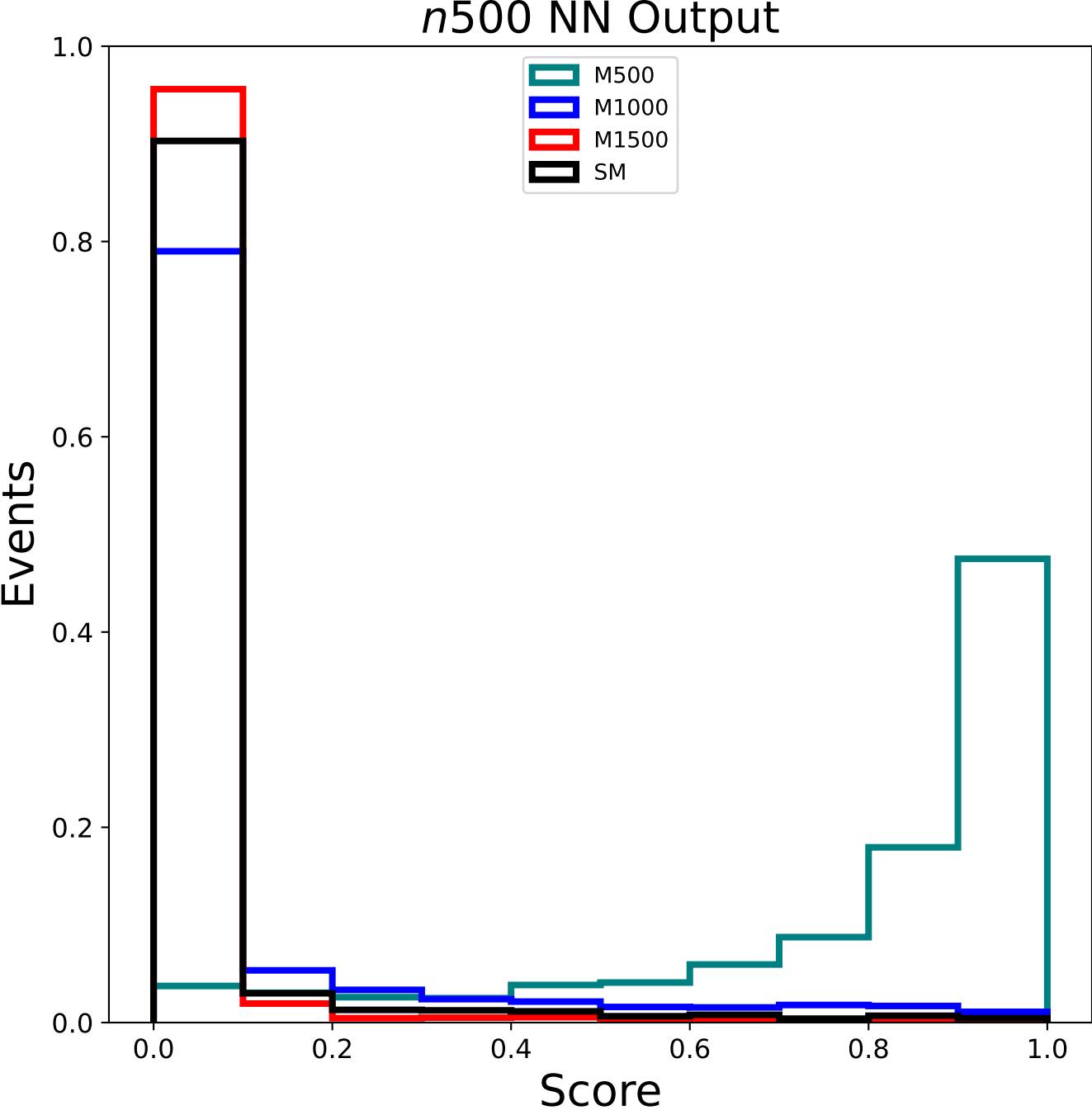

1.0
0.8
0.6
0.4
0.2
0.0
Signal Efficiency
0.0
0.2
0.4
0.6
0.8
1.0
Background Efficiency
n500 Training (AUC = 0.978)
n500 Validation (AUC = 0.977)
n1000 Training (AUC = 0.956)
n1000 Validation (AUC = 0.947)
n1500 Training (AUC = 0.977)
n1500 Validation (AUC = 0.974)
nSM Training (AUC = 0.997)
nSM Validation (AUC = 0.997)

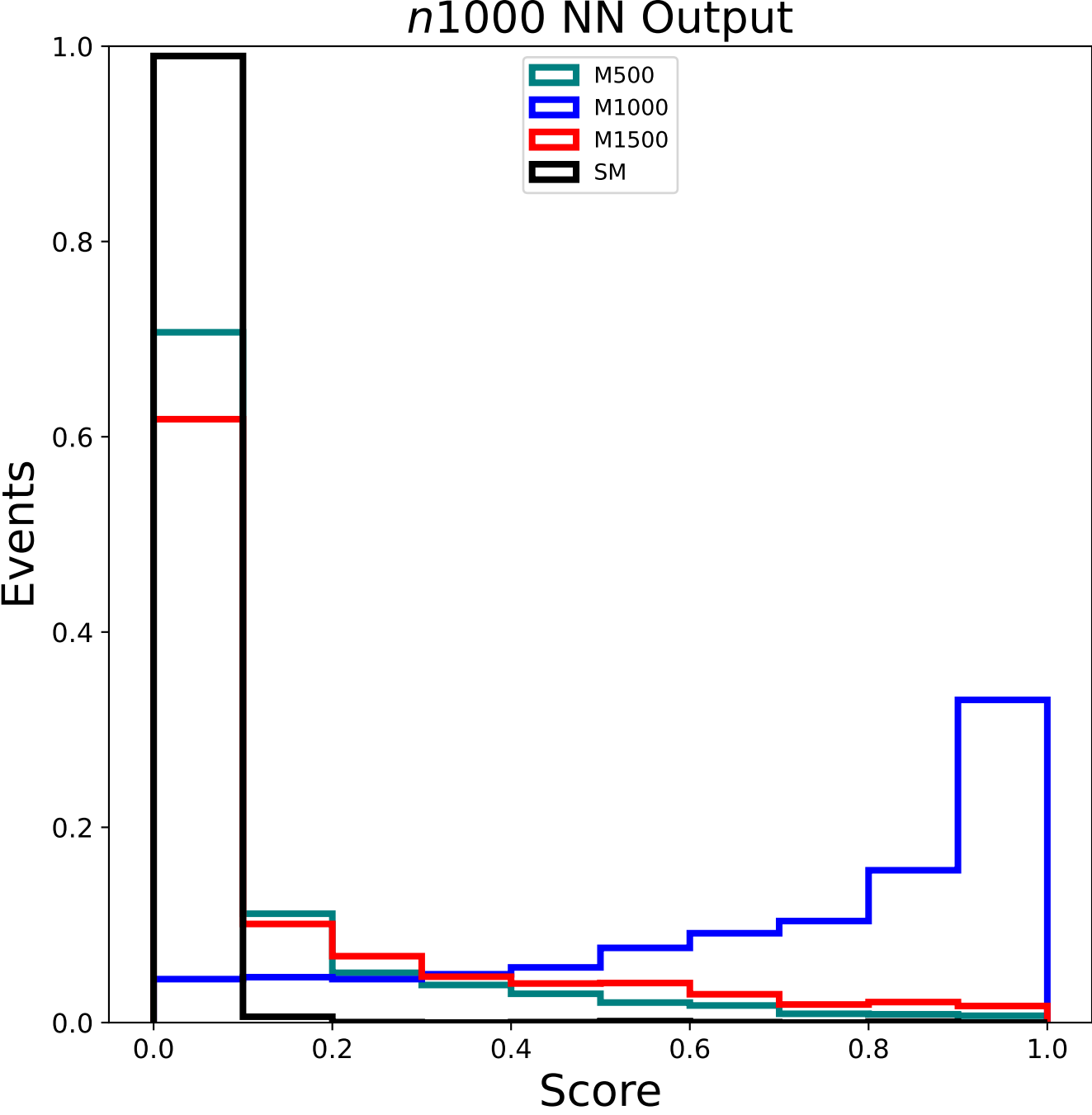

NN Output for procM500
n500
n1000
n1500
nSM
Events
Score
0.0
0.2
0.4
0.6
0.8
1.0

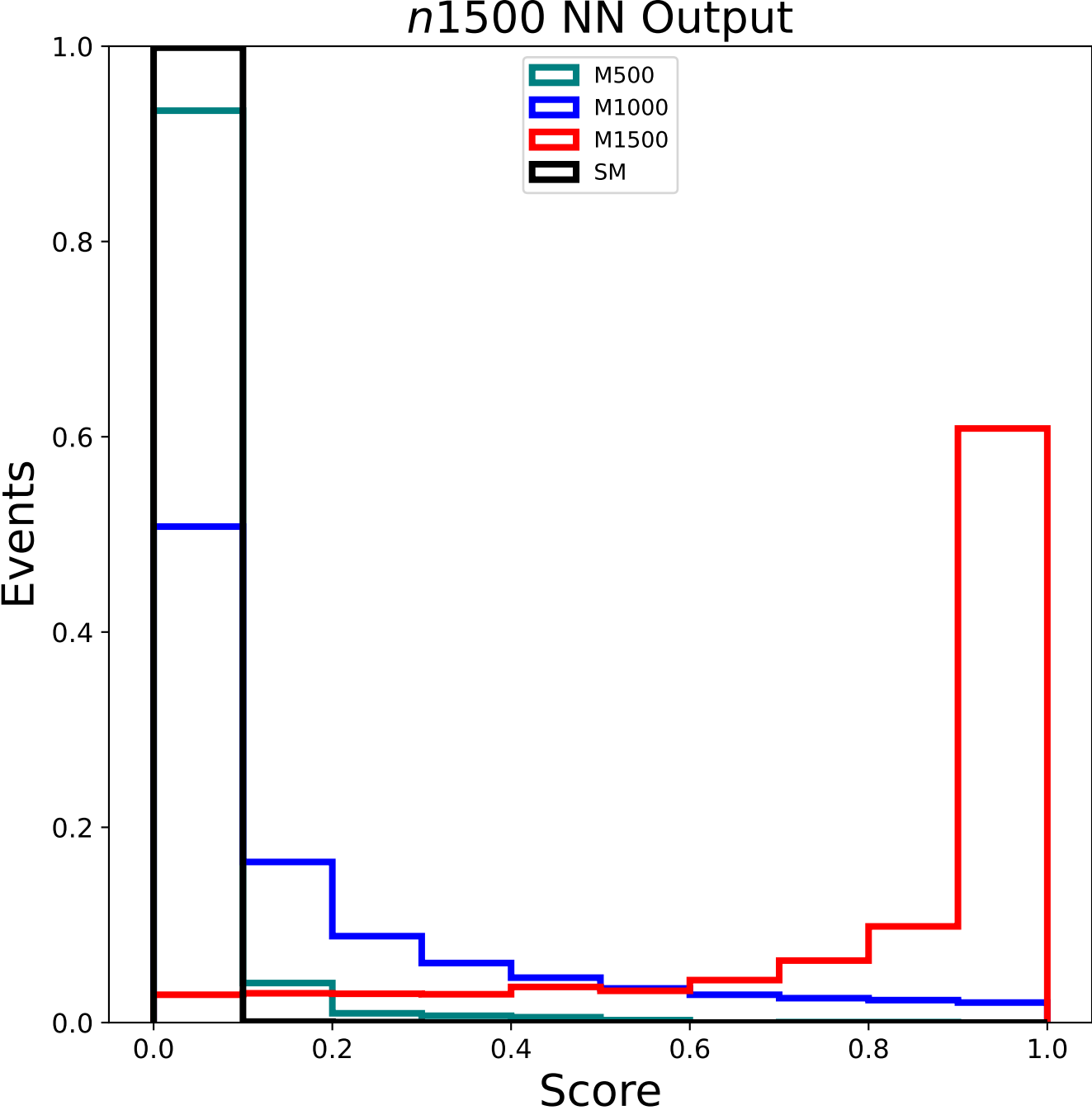

NN Output for procM1000
n500
n1000
n1500
nSM
Events
Score
0.0
0.2
0.4
0.6
0.8
1.0

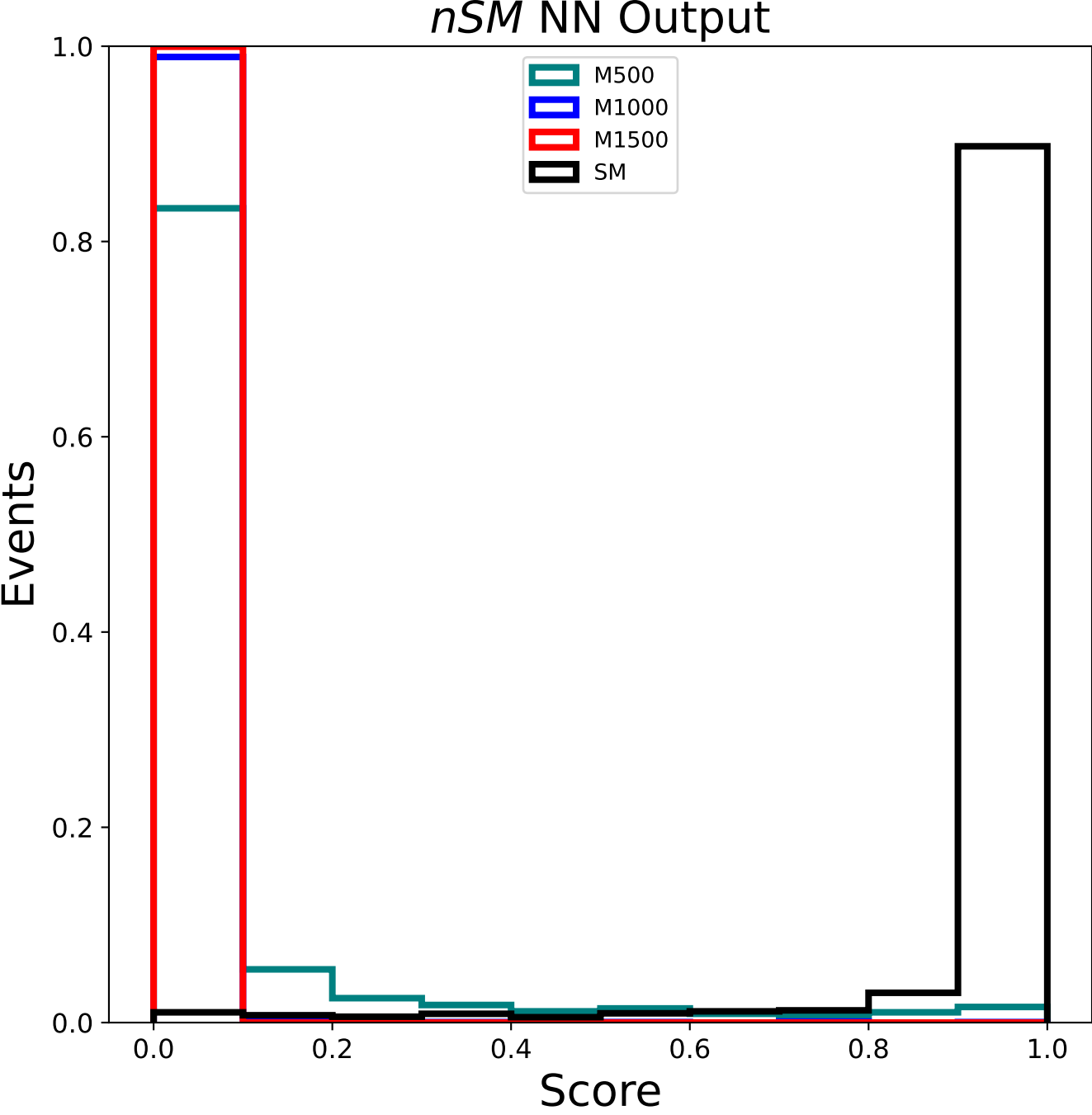

NN Output for procM1500
n500
n1000
n1500
nSM
Events
Score
0.0
0.2
0.4
0.6
0.8
1.0

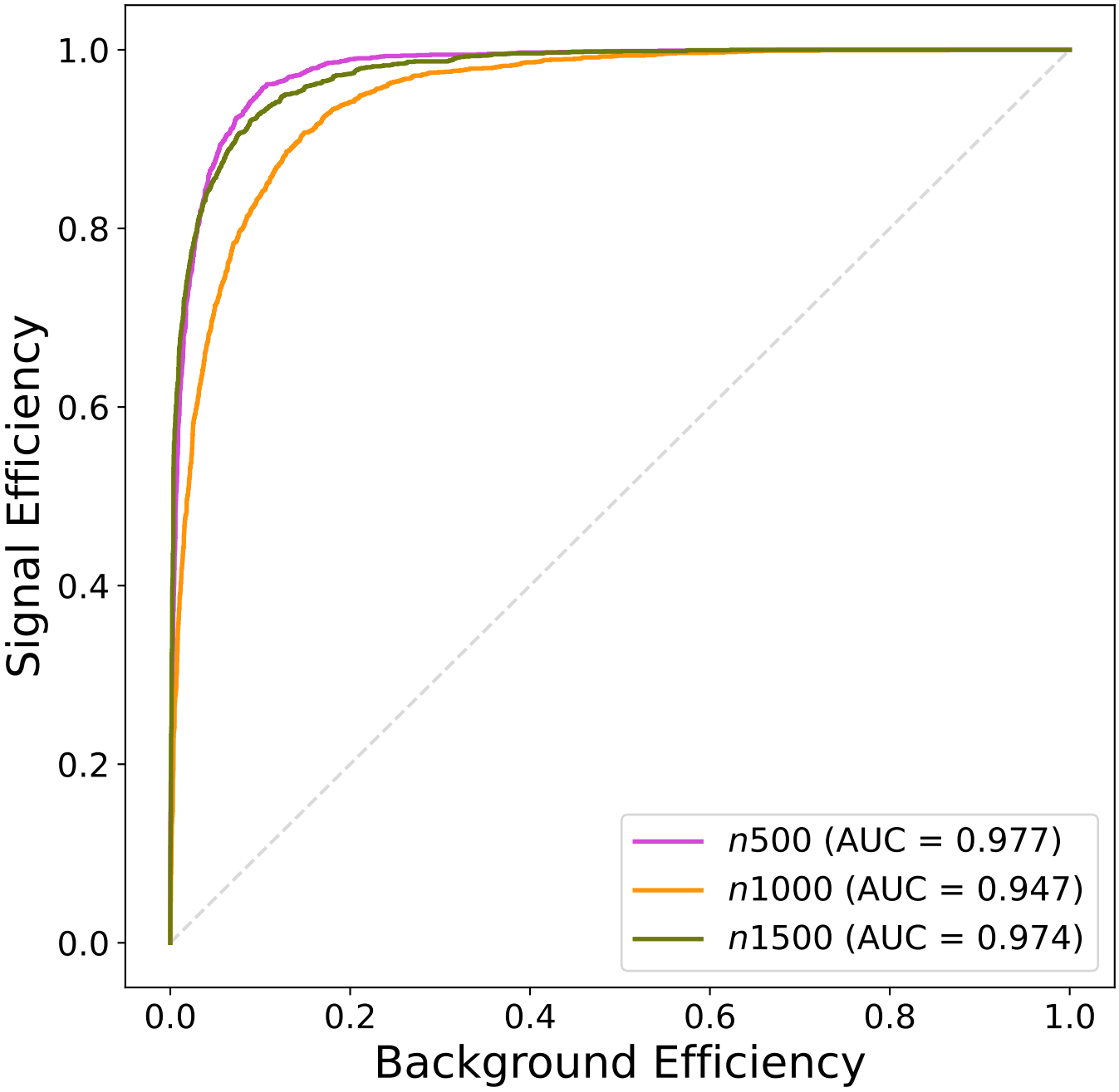

NN Output for procBkg
n500
n1000
n1500
nSM
Events
Score
0.0
0.2
0.4
0.6
0.8
1.0


n500 NN Output
M500
M1000
M1500
SM
Events
Score
0.0
0.2
0.4
0.6
0.8
1.0


n1000 NN Output
M500
M1000
M1500
SM
Events
Score
0.0
0.2
0.4
0.6
0.8
1.0


n1500 NN Output
M500
M1000
M1500
SM
Events
Score
0.0
0.2
0.4
0.6
0.8
1.0


nSM NN Output
M500
M1000
M1500
SM
Events
Score
0.0
0.2
0.4
0.6
0.8
1.0


1.0
0.8
0.6
0.4
0.2
0.0
Signal Efficiency
0.0
0.2
0.4
0.6
0.8
1.0
Background Efficiency
n500 (AUC = 0.977)
n1000 (AUC = 0.947)
n1500 (AUC = 0.974)